

As P and Q coincide, the ratio of Qv to Qx becomes a ratio of equality by Cor. 2 of Lemma VII, so $Qx^2 \rightarrow 4PS \cdot QR$. From the similar triangles QxT and SPN , $Qx^2 : QT^2 = PS^2 : SN^2 = PS : SA$, whence $QT^2 \propto SP \cdot QR$. By Cor. 1 of Prop. VI the force is measured as $1/(SP^2 \cdot QT^2/QR)$, and substituting gives the **inverse-square law** $F \propto 1/SP^2$. The parabola obeys the same law as the ellipse. *Q.E.D.*